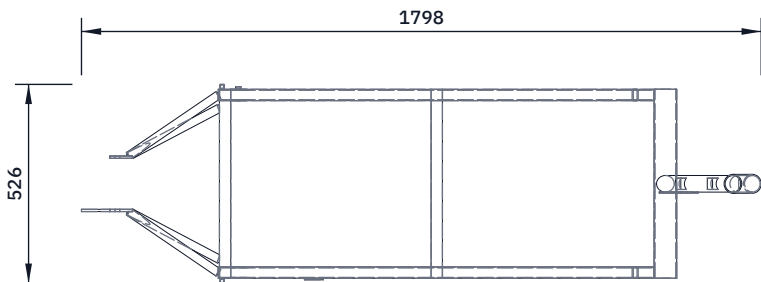
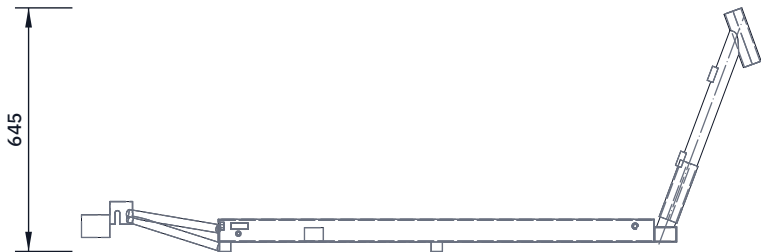


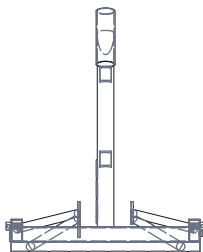
REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-01	AC



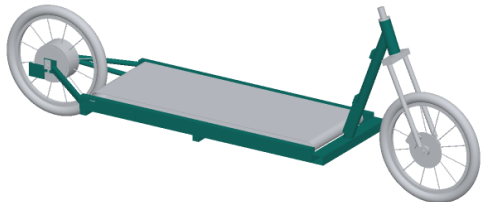
TOP VIEW
Scale 1:20; looking down (along -Z)



FRONT VIEW
Scale 1:20; looking at the front (along +Y)



RIGHT VIEW
Scale 1:20; looking at the right side (along -X)



WHERE IT GOES
This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Weld on a flat table in this order, tacking everything first:

1. Rails on edge, parallel, 440 mm apart inside; cross members under them; nose beam across their front ends. Check the diagonals agree within 2 mm, then weld.
2. Head tube held in a jig at 70 degrees, its bottom end 640 mm above the table plane under the wheels (470 mm above the rails' underside), 1410 mm ahead of the rails' rear ends, on the centre line.
3. Down tube between the nose beam and the head tube (sheet 103).
4. Dropouts on a dummy 135 mm axle, axle 77 mm above the rails' underside and 270 mm behind their rear ends; then the four stays.
5. End caps, lugs, plates, tabs and nuts (sheet 105).

Weld in short runs, alternating sides, to limit distortion.

Check: rails flat and parallel within 2 mm; head tube on the centre line within 1 mm; rear axle square to the centre line; then paint.

STEPGEN

github.com/BoujeeEnjinia1701/stepgen

StepGen main frame: weld-up sketch

DRAWING NO. SGN-DWG-106			REV P1	SHEET 1 of 1	SIZE ANSI B
SCALE 1:20	UNITS mm	PROJECTION 	DATE 2026-10-01		DRAWN Amish Chadha
MATERIAL / NOTES Welded mild steel, S235 class; paint after welding					
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